

# ECE 5159-Class 3

## DIGITAL SYSTEM DESIGN WITH VHDL



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# CAD TOOLS

- The approach used so far can only be used by a designer manually for small circuits.
- However, logic circuits found in complex systems are designed using sophisticated CAD tools that automatically implement the synthesis techniques.
- To design a logic circuit, a number of CAD tools are needed. They are usually packaged together into a *CAD system*, which typically includes tools for the following tasks:

# CAD TOOLS Contd

- Design entry
- Synthesis and optimization
- Simulation
- Physical design .

# CAD TOOLS Contd

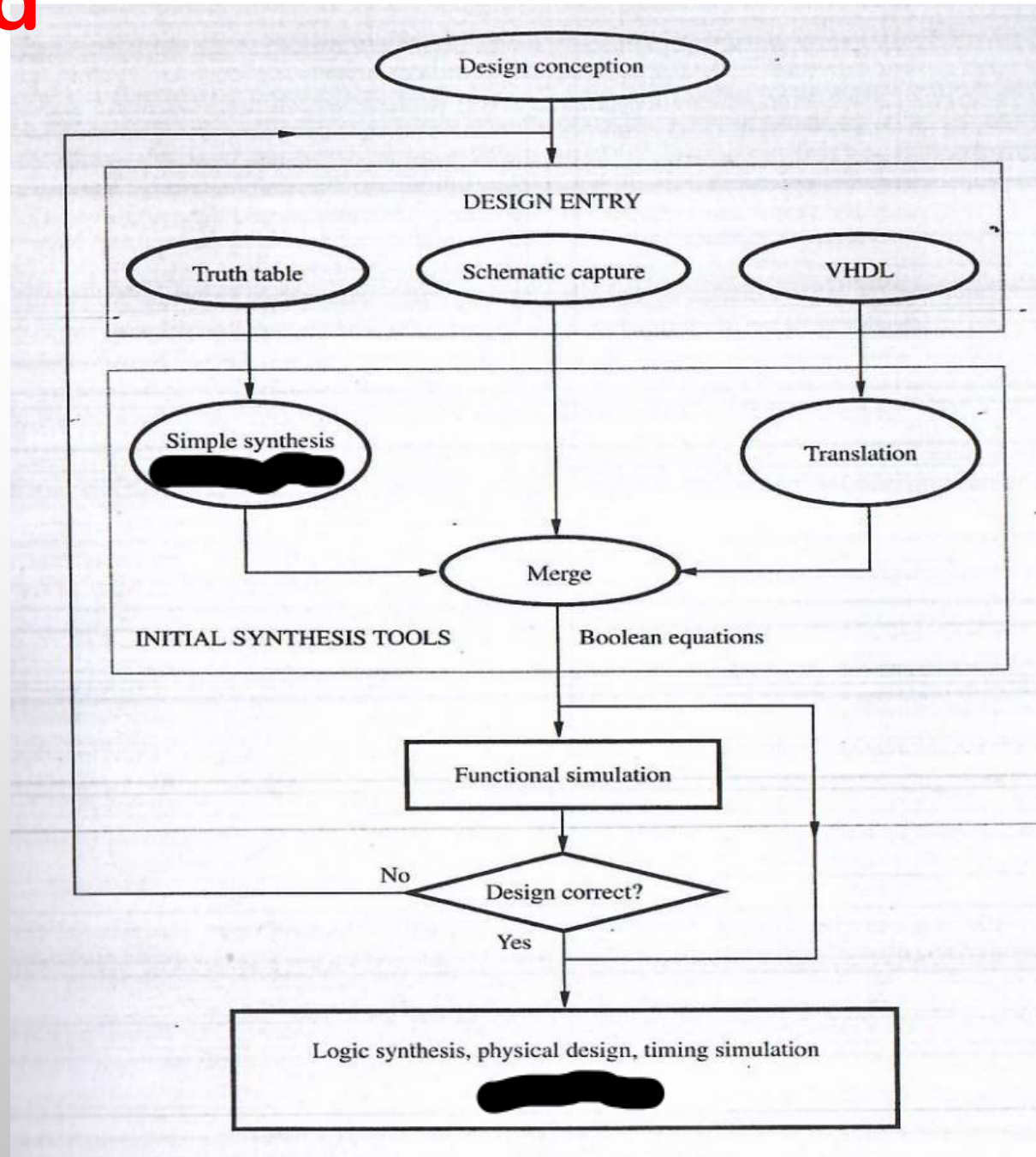


Figure 1. Framework of a Typical CAD System

# DESIGN ENTRY

Design entry is the stage used to enter the description of the circuit being designed.

There are three Design entry methods:

- Truth Table / Timing Diagram
- Schematic capture
- Writing source code in a hardware description language

# DESIGN ENTRY: Truth Table

We are familiar with Truth Table and timing diagram using waveform e.g.

| X <sub>1</sub> | X <sub>2</sub> | X <sub>3</sub> | F |
|----------------|----------------|----------------|---|
| 0              | 0              | 0              | 0 |
| 0              | 0              | 1              | 1 |
| 0              | 1              | 0              | 1 |
| 0              | 1              | 1              | 0 |
| 1              | 0              | 0              | 1 |
| 1              | 0              | 1              | 0 |
| 1              | 1              | 0              | 0 |
| 1              | 1              | 1              | 1 |

Figure 2. Truth Table

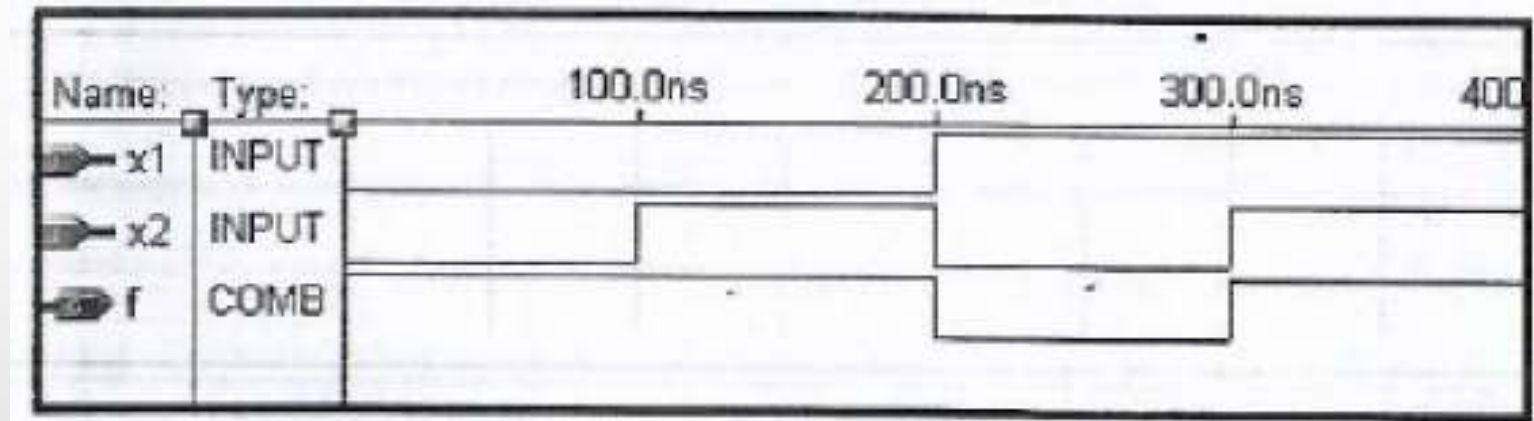


Figure 3. Waveform Editor



# DESIGN ENTRY: Truth Table

Corresponding Truth Table for the waveform

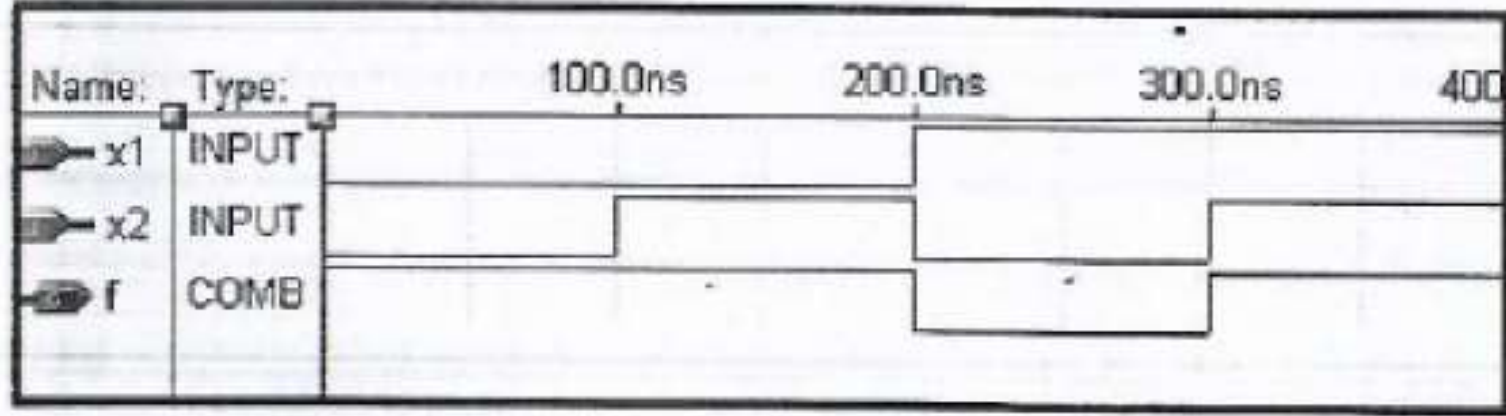


Figure 3. Waveform Editor

| $X_1$ | $X_2$ | F |
|-------|-------|---|
| 0     | 0     | 1 |
| 0     | 1     | 1 |
| 1     | 0     | 0 |
| 1     | 1     | 1 |

Figure 3b. Truth Table

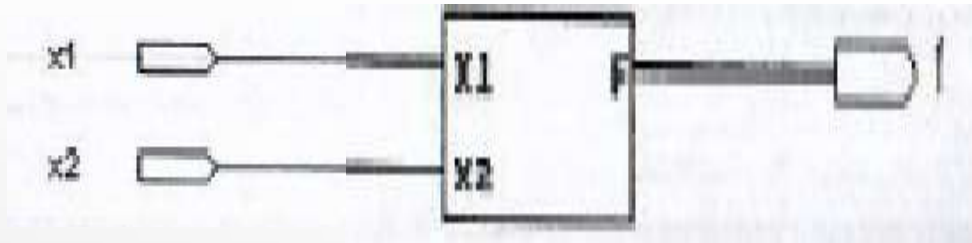
$F = ?$

# SCHEMATIC CAPTURE

- A schematic capture tool uses the graphics capabilities of a computer and a computer mouse to allow the user to draw a schematic diagram.
- There is library of various types of gates with different numbers of inputs for inclusion in the schematic
- The tool provides a graphical way of interconnecting the gates to create a logic network.
- Any subcircuits that have been previously created, using either different design entry methods or the schematic capture tool itself, can be represented as graphical symbols and included in the schematic, this method is known as hierarchical design.



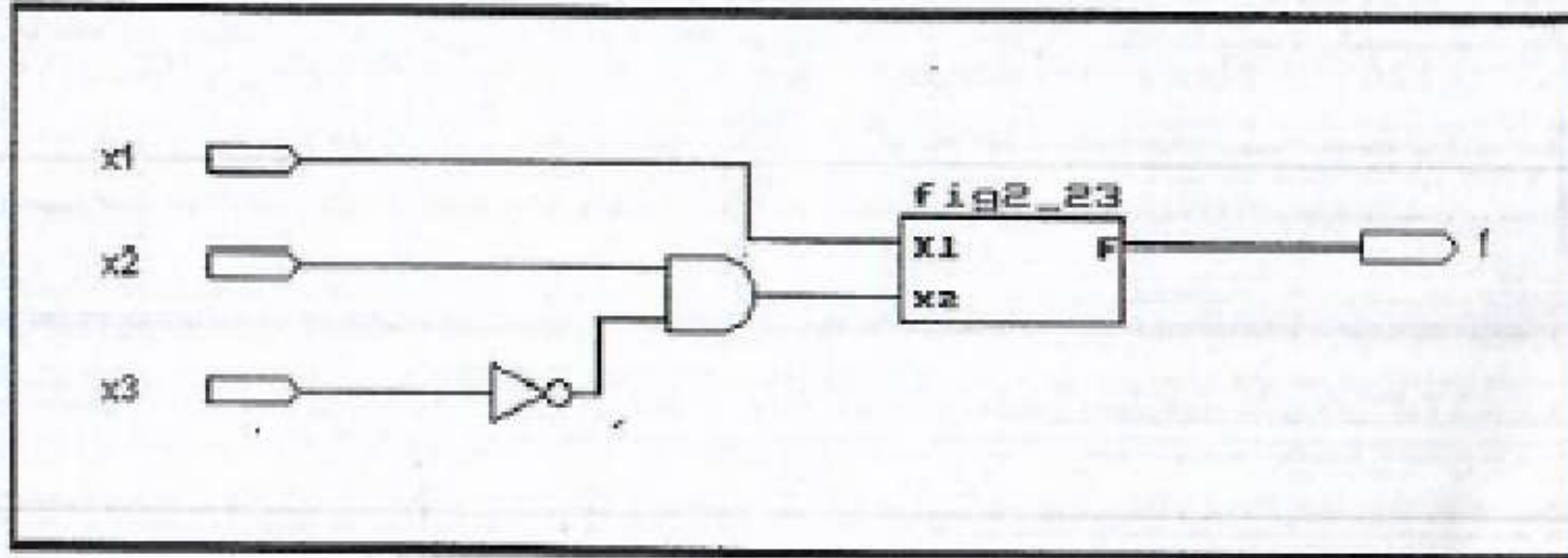
# SCHEMATIC CAPTURE Contd



$F = ?$

| $X_1$ | $X_2$ | $F$ |
|-------|-------|-----|
| 0     | 0     | 1   |
| 0     | 1     | 1   |
| 1     | 0     | 0   |
| 1     | 1     | 1   |

Figure 4a. Schematic capture on the Graphic Editor



$F = ?$

Figure 4b. Schematic capture of hierarchical design on the Graphic Editor

# HARDWARE DESCRIPTION LANGUAGE

- A hardware description language (HDL) is similar to a typical computer programming language except that an HDL is used to describe hardware rather than a program to be executed on a computer.
- Many commercial HDLs are available.
  - Some are proprietary,
  - Some supported by virtually all vendors that provide digital hardware technology and is officially endorsed by *IEEE* as standard.

# HARDWARE DESCRIPTION LANGUAGE (HDL)

There are two HDL

- Verilog
- Very high speed integrated circuit Hardware Description Language (VHDL)

## Advantages of VHDL

VHDL provides **design portability**.

A circuit specified in VHDL can be implemented in different types of chips and with CAD tools provided by different companies, without having to change the VHDL specification.

VHDL encourages **sharing and reuse** of VHDL-described circuits

# HARDWARE DESCRIPTION LANGUAGE (HDL)

Design entry of a logic circuit is done by writing VHDL code.

Signals in the circuit are represented as variables in the source code, and logic functions are expressed by assigning values to these variables

## SYNTHESIS AND OPTIMIZATION

- CAD Synthesis is the process of generating a logic circuit from a truth table.
- CAD Synthesis tools perform this process automatically.
- CAD Synthesis tools also handle many other tasks: e.g. *translating*, or *compiling*, VHDL code into a network of logic gates

# SYNTHESIS AND OPTIMIZATION

Regardless of what type of design entry is used, the initial logic expressions produced by the synthesis tools are not likely to be in an optimal form.

One of the most important task of the synthesis tools is to manipulate the user 's design to automatically produce an equivalent but better circuit. This step of synthesis is called *logic synthesis* , or *logic optimization*

# SIMULATION

Once the design entry and synthesis are complete, it is useful to verify that the designed circuit functions as expected. The tool that performs this task is called a **functional simulator**

## Stages of Simulation

- The user's initial design is represented by the logic equations generated during synthesis.
- the user specifies valuations of the circuit's inputs that should be applied to these equations during simulation.
- For each valuation, the simulator evaluates the outputs produced by the equations.
- The output of the simulation is provided either in truth-table form or as a timing diagram.
- The user examines this output to verify that the-circuit operates as required.



# INTRODUCTION TO VHDL

VHDL was developed as a part of the effort for the rapid advances in integrated circuit technology which led to develop standard design practices for digital circuits in the 1980s

VHDL has become the industry standard language for describing digital circuits, largely because it is an official IEEE standard.

The original standard for VHDL was adopted in 1987 and called IEEE 1076. A revised standard was adopted in 1993 and called IEEE 1164.

# INTRODUCTION TO VHDL

VHDL was originally intended to serve two main purposes.

- used as a documentation language for describing the structure of complex digital circuits
- provided features for modeling the behavior of a digital circuit, which allowed its use as input to software programs that were then used to simulate the circuit's operation

# INTRODUCTION TO VHDL

In recent years VHDL is now used for

Documentation

Simulation

Design entry in CAD systems.

The CAD tools are used to synthesize the VHDL code into a hardware implementation of the described circuit.

# REPRESENTING SIGNALS IN VHDL

Consider the circuit of Figure 5

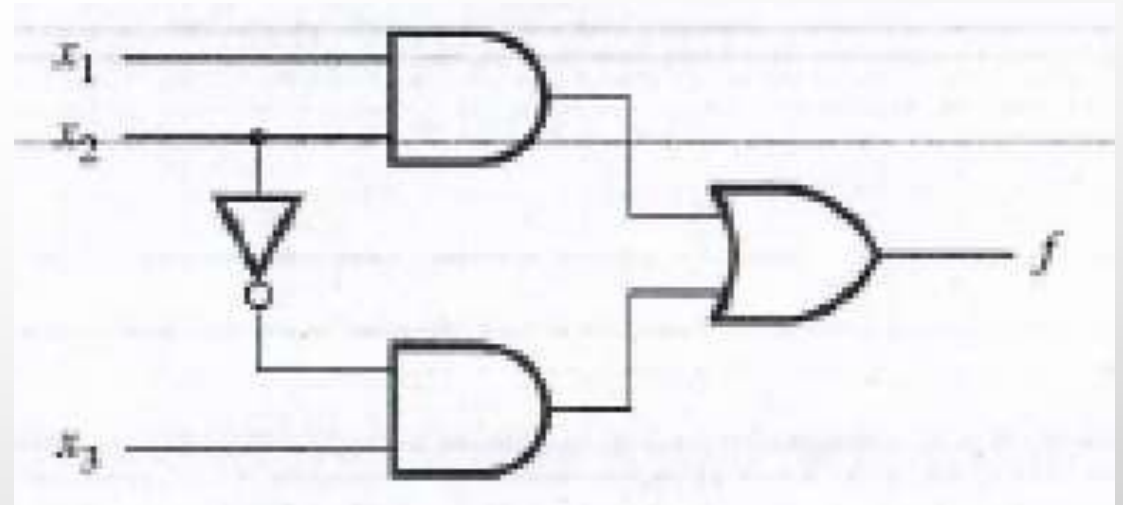


Figure 5. Schematic

```
ENTITY example1 IS
    PORT (x1, x2, x3: IN      BIT;
          f      : OUT BIT);
END example1;
```

# REPRESENTING SIGNALS IN VHDL

## Steps Involves

- Declare the input and output signals. This is done using a construct called an **entity**. An entity must have a name, in this case example1

The input and output signals for the entity are called its *ports* and they are identified by the word **PORT**.

Each port has an associated mode that specifies whether it is an input (IN) to the entity or an output (OUT) from the entity.

Each port represents a signal hence it has an associated type, in this case type **BIT**.

# REPRESENTING SIGNALS IN VHDL

## Steps Involves continued

- The circuit's functionality is specified with a VHDL construct called an **ARCHITECTURE**.
- An architecture must be given a name e.g. *LogicFunc*.
- Architecture specifies the functionality of the design using a logic expression.
- VHDL has built-in support for the following Boolean operators: AND, OR, NOT, NAND, NOR, XOR and XNOR.
- Following the BEGIN keyword. The architecture specifies, using the VHDL signal assignment operator `<=`, that output *f* should be assigned the result of the logic expression on the right-hand side of the operator.



# REPRESENTING SIGNALS IN VHDL

## Circuit Functionality

For the expression

$$f \leq x_1 \text{ AND } x_2 \text{ OR NOT } x_2 \text{ AND } x_3$$

## VHDL code for it

$$f \leq (x_1 \text{ AND } x_2) \text{ OR } (\text{NOT } x_2 \text{ AND } x_3)$$

## NOT

$$f \leq (x_1 \text{ AND } x_2) \text{ OR } ((\text{NOT } x_2) \text{ AND } x_3)$$

# REPRESENTING SIGNALS IN VHDL

## •Circuit Functionality

ARCHITECTURE LogicFunc OF example1 IS  
BEGIN

$f \leq (x_1 \text{ AND } x_2) \text{ OR } (\text{NOT } x_2 \text{ AND } x_3);$   
END LogicFunc;

# REPRESENTING SIGNALS IN VHDL

Complete VHDL code illustrates that a VHDL source code file has two main sections: an entity and an architecture.

```
ENTITY example1 IS
```

```
    PORT (x1, x2, x3: IN    BIT ;  
          f      : OUT BIT) ;
```

```
END example1 ;
```

```
ARCHITECTURE LogicFunc OF example1 IS  
BEGIN
```

```
    f <= (x1 AND x2) OR (NOT x2 AND x3) ;  
END LogicFunc ;
```

# CLASS WORK 1

Write a Complete VHDL code for the circuit below

